Centrality Crowding in Asset Prices: A Limited-Attention Network Model*

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Abstract

This paper develops a one-period behavioral model of price aggregation in social networks. Investors combine noisy private signals with a centrality-weighted social feed. In the equal-precision benchmark, pricing error is summarized by $M = \sigma^2 H$, where H is the Herfindahl concentration of equilibrium signal weights. Greater centrality amplification increases H in non-uniform networks, reducing price informativeness. A Gaussian projection implies larger one-period corrections when pricing error is greater. With heterogeneous precision, centrality-based attention is efficient if and only if visibility weights align with precision weights. The model links network concentration to fragile price discovery.

Keywords: social networks; limited attention; asset prices; price discovery; centrality; information aggregation
JEL classification: D83; D85; G12; G14; G41

1 Introduction

Financial markets increasingly process information through social structures. Retail investors follow influencers and discussion platforms; professional investors exchange views through education, employment, board, and geographic networks; and digital platforms rank financial content according to engagement. A large empirical literature documents that social interaction affects participation, portfolio choice, and the speed with which investors respond to information (Shiller and Pound, 1989; Hong et al., 2004, 2005; Ivković and Weisbenner, 2007; Kuchler and Stroebel, 2021; Hirshleifer et al., 2025). These mechanisms can accelerate diffusion, but diffusion and diversification are not the same thing. If many investors repeatedly attend to the same highly visible source, communication can make the information set socially redundant even when every transmitted signal is truthful.

This paper isolates that redundancy channel in a deliberately parsimonious behavioral model. Each investor observes an independent noisy signal about a common payoff. Trading beliefs combine the investor's own signal with a social statistic whose weights are determined by network centrality. A parameter ϕ governs how aggressively a platform or communication environment amplifies central nodes. When ϕ rises, a non-uniform centrality vector is transformed into a

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more concentrated attention vector. Competitive market clearing then maps this attention concentration directly into concentration of the signal weights embedded in price.

The model should be interpreted as a limited-attention aggregation model rather than as a noisy rational-expectations equilibrium. Investors do not learn from the equilibrium price; the perceived residual payoff variance entering demand is an exogenous reduced-form parameter; net asset supply is fixed rather than stochastic; and centrality-based weights capture visibility rather than Bayesian precision. These distinctions matter. Standard noisy rational-expectations models study endogenous inference from prices under dispersed information and often stochastic supply (Grossman, 1976; Grossman and Stiglitz, 1980; Hellwig, 1980; Diamond and Verrecchia, 1981). Here the question is narrower: conditional on a reduced-form rule that overweights socially visible signals, how does the concentration of those weights affect price discovery?

The main result is a sufficient-statistic representation. Let a_i denote the equilibrium weight of investor i's private signal in the risk-premium-adjusted price and let

$$H = \sum_{i=1}^{n} a_i^2.$$

With independent equal-precision signal errors of variance σ^2, mean squared pricing error is

$$M \equiv \mathbb{E}\big[(\widetilde{p} - \theta)^2\big] = \sigma^2 H.$$

Thus the network affects price discovery only through the concentration of equilibrium signal weights. Proposition 2 shows that centrality amplification ϕ increases H whenever network centrality is heterogeneous and social information receives positive weight. This comparative static is the model's centrality-specific mechanism.

The paper is closely related to DeMarzo et al. (2003). DeMarzo, Vayanos, and Zwiebel develop a boundedly rational model of repeated opinion exchange in which agents fail to correct for repeated information; as a result, social influence depends on network connectedness as well as accuracy. The present model does not reproduce their repeated-averaging dynamics. Instead, it takes centrality-dependent visibility as a one-shot attention rule and asks how that rule is transmitted into a market price. The marginal contribution is therefore threefold. First, it provides the closed-form market-price sufficient statistic $M = \sigma^2 H$, which translates social concentration into a directly interpretable price-discovery loss. Second, it derives a platform-style comparative static in the amplification parameter ϕ, distinguishing network architecture from the intensity with which centrality is amplified. Third, with heterogeneous signal precision, it characterizes the exact centrality–precision alignment condition under which centrality-based attention is efficient. These results generate empirical predictions about observed attention concentration, price informativeness, and the moderating role of source accuracy.

The static model also admits a simple Gaussian signal-extraction interpretation. If the current risk-adjusted price is viewed as a noisy signal of the terminal payoff, the expected one-step correction toward the payoff is larger when M is larger. This observation is intentionally modest: stronger mean reversion in a noisier Gaussian signal is generic and is not a distinct dynamic prediction of the network model. The model-specific content is the link from centrality amplification to M and, under heterogeneous precision, the distance between centrality weights and precision-efficient weights.

The remainder of the paper is organized as follows. Section 2 presents the behavioral aggregation model and derives the sufficient statistic. Section 3 illustrates the centrality

mechanism. Section 4 establishes the centrality-amplification comparative static and gives its one-period empirical interpretation. Section 5 separates social attention intensity from attention concentration. Section 6 introduces heterogeneous precision and derives the precision-alignment condition. Section 7 concludes. Proofs are collected in the Appendix.

2 Networked Attention and Trading

2.1 Signals and social attention

There are $n \geq 2$ risk-averse investors, indexed by $i, j \in \{1, \ldots, n\}$, and one risky asset. The terminal payoff is

$$\theta \sim N(0, \sigma_\theta^2). \tag{1}$$

Here θ is the asset's terminal fundamental payoff. Its prior mean is normalized to zero, while $\sigma_\theta^2 > 0$ measures fundamental uncertainty : a larger σ_θ^2 means that the payoff is more dispersed before any private information is observed. This variance is conceptually distinct from the signal-noise variance introduced below.

Investor i observes a private signal

$$x_i = \theta + \varepsilon_i, \qquad \varepsilon_i \overset{iid}{\sim} N(0, \sigma^2), \tag{2}$$

where x_i is investor i's noisy observation of the common payoff, ε_i is her idiosyncratic signal error, and $\sigma^2 > 0$ is the common signal-error variance in the equal-precision benchmark. Thus lower σ^2 means more precise private information. The errors are independent across investors and independent of θ. This independence is important: any redundancy in the price comes from repeated social weighting rather than mechanically correlated signal errors.

Let the social network be summarized by a strictly positive normalized centrality vector $c = (c_1, \ldots, c_n)$, with $c_i > 0$ and $\sum_i c_i = 1$. The scalar c_i is investor i's normalized social centrality or visibility share: larger c_i means that investor i occupies a more prominent position in the communication network. The normalization is without loss of generality because only relative centrality matters. The underlying centrality measure can be eigenvector centrality or another positive measure of network visibility. A social feed transforms centrality into attention weights according to

$$r_i(\phi) = \frac{c_i^\phi}{\sum_{j=1}^n c_j^\phi}, \qquad \phi \geq 0. \tag{3}$$

Here $r_i(\phi) \in (0, 1)$ is investor i's share of total social attention, and the denominator normalizes the weights so that $\sum_i r_i(\phi) = 1$. The parameter ϕ is the centrality-amplification parameter: it controls how strongly differences in network centrality are converted into differences in visibility. At $\phi = 0$, centrality is ignored and $r_i = 1/n$ for every investor. At $\phi = 1$, attention is proportional to normalized centrality, so $r_i = c_i$. Values $0 < \phi < 1$ compress centrality differences, whereas $\phi > 1$ disproportionately amplifies the most central nodes. Hence ϕ changes the concentration of attention without changing the underlying network vector c.

Investor i forms the reduced-form trading belief

$$b_i = (1 - \rho)x_i + \rho \sum_{j=1}^n r_j(\phi) x_j, \qquad \rho \in [0, 1], \tag{4}$$

where b_i is investor i's perceived payoff index used for trading and is measured in the same payoff units as θ and x_i. The parameter ρ is the social-attention intensity: $1-\rho$ is the weight on investor i's own private signal, while ρ is the weight on the common social statistic $\sum_j r_j(\phi)x_j$. Thus $\rho = 0$ corresponds to purely private-information trading and $\rho = 1$ to complete reliance on the social feed. Importantly, ρ governs the amount of social reliance, whereas ϕ governs the concentration of that social attention across sources. The social statistic contains truthful signals, but it need not weight them by statistical precision. The distortion studied below is therefore an attention distortion rather than a misreporting distortion.

2.2 Behavioral interpretation and scope

Four modeling choices delimit the interpretation of the analysis. First, investors do not infer information from the equilibrium price. Equation (4) is a reduced-form behavioral belief rule, not a Bayesian posterior conditioned on prices. Second, the residual payoff variance σ_u^2 entering demand below is perceived and exogenous; it is not solved endogenously from the equilibrium information structure. Third, aggregate net supply Q is a fixed deterministic quantity throughout the paper. There are no noise traders or stochastic supply shocks. Fourth, $r_i(\phi)$ reflects social visibility. In the baseline equal-precision case visibility and precision are conceptually distinct even though all σ^2 are identical; Section 6 allows precision to differ explicitly.

These assumptions are chosen to expose the centrality-concentration channel in closed form. They also imply that the model is not a fully rational expectations model. The competitive pricing step below aggregates behavioral beliefs through market clearing, but it does not make the belief formation rule itself rational-expectations consistent.

2.3 Demand, market clearing, and the sufficient statistic

Investors have CARA preferences and use a common perceived residual payoff variance $\sigma_u^2 > 0$. Given belief index b_i, demand is

$$d_i = \frac{b_i - p}{\gamma \sigma_u^2}, \tag{5}$$

where d_i is investor i's desired position in the risky asset and p is its current market price. The parameter $\gamma > 0$ is the common coefficient of absolute risk aversion: a larger γ makes investors less willing to absorb a given perceived payoff-price gap. The parameter $\sigma_u^2 > 0$ is investors' common perceived residual payoff variance; it is an exogenous reduced-form risk parameter rather than an endogenous posterior variance. Accordingly, $\gamma\sigma_u^2$ is the effective risk-bearing coefficient in the demand schedule. Market clearing requires $\sum_i d_i = Q$, where Q is the fixed deterministic net supply of the risky asset. A larger positive Q requires investors collectively to hold more risk and therefore generates a larger equilibrium risk-premium discount.

Proposition 1 (Linear price and sufficient statistic)**.** *The unique market-clearing price is*

$$p = \sum_{i=1}^{n} a_i(\rho,\phi)x_i - \lambda Q, \qquad a_i(\rho,\phi) = \frac{1-\rho}{n} + \rho r_i(\phi), \qquad \lambda = \frac{\gamma\sigma_u^2}{n}, \tag{6}$$

with $\sum_i a_i = 1$. The coefficient $a_i(\rho,\phi)$ is the equilibrium loading of price on investor i's private signal. Its first component, $(1-\rho)/n$, comes from averaging investors' own-signal components, while its second component, $\rho r_i(\phi)$, comes from the common social feed. The coefficient $\lambda \equiv \gamma\sigma_u^2/n > 0$ is the price discount per unit of fixed net supply, so λQ is a

deterministic risk-premium term. In particular, λ depends on risk aversion, perceived residual risk, and the number of risk-bearing investors, but not on the information network.

Define the risk-premium-adjusted price $\widetilde{p} \equiv p + \lambda Q$. The variable $\widetilde{p}$ removes the deterministic supply discount and therefore isolates the informational component of the market price. Then

$$\widetilde{p} - \theta = \sum_{i=1}^{n} a_i \varepsilon_i, \tag{7}$$

so the pricing error is exactly the weighted average of idiosyncratic signal errors that survives market aggregation. Mean squared pricing error is

$$M(\rho, \phi) \equiv \mathbb{E}\big[(\widetilde{p} - \theta)^2\big] = \sigma^2 H(\rho, \phi), \tag{8}$$

where $M(\rho, \phi)$ is the ex ante mean squared deviation of the informational price from the true payoff. Because the pricing error has mean zero, M is also its variance. The term σ^2 measures the noise of each underlying private signal, whereas $H(\rho, \phi)$ measures how strongly those errors are concentrated by the equilibrium weighting scheme. Specifically,

$$H(\rho, \phi) = \sum_{i=1}^{n} a_i^2 = \frac{1}{n} + \rho^2 \left(R(\phi) - \frac{1}{n} \right), \qquad R(\phi) = \sum_{i=1}^{n} r_i(\phi)^2. \tag{9}$$

Here $H \in [1/n, 1]$ is the Herfindahl concentration index of the equilibrium price weights a_i, while $R(\phi) \in [1/n, 1]$ is the Herfindahl concentration index of the social-attention weights $r_i(\phi)$. The lower bound $1/n$ corresponds to equal weighting across all investors; values closer to one indicate that the price or the social feed relies on only a few signals. The identity also shows why ρ enters quadratically: social concentration affects pricing error only to the extent that the social component is loaded into equilibrium price weights.

Proof. See Appendix. □

The result separates the scale of signal noise, σ^2, from the concentration of the weights that survive aggregation, H. The product $M = \sigma^2 H$ is sufficient for all one-period pricing-error implications below. In particular,

$$\mathrm{Var}(\widetilde{p} \mid \theta) = M, \qquad \mathrm{Var}(\widetilde{p}) = \sigma_\theta^2 + M. \tag{10}$$

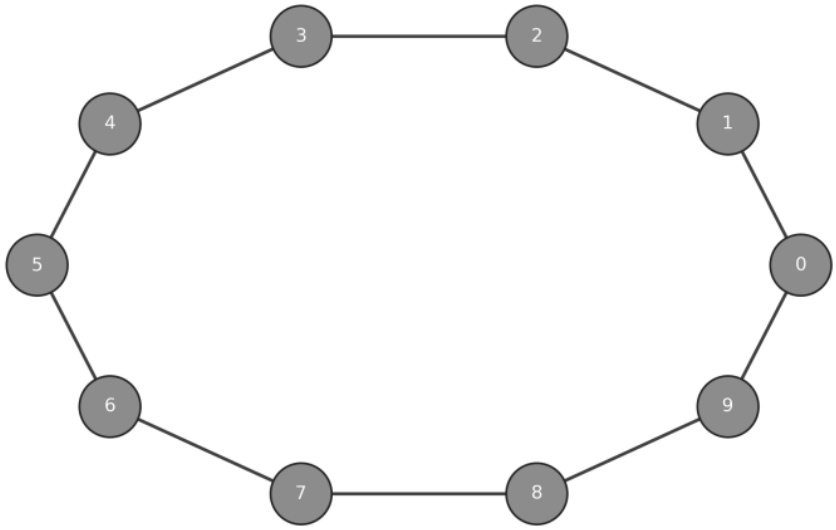
The first object, $\mathrm{Var}(\widetilde{p} \mid \theta) = M$, holds the fundamental fixed and therefore measures purely non-fundamental price dispersion generated by noisy signals. The second object, $\mathrm{Var}(\widetilde{p}) = \sigma_\theta^2 + M$, combines fundamental payoff variation σ_θ^2 with pricing noise M. Accordingly, the paper uses "pricing noise" or "pricing-error variance" rather than interpreting M as a fully fledged dynamic excess-volatility statistic.

Equation (9) also clarifies why communication per se need not be harmful. The idiosyncratic component $(1 - \rho)/n$ is diversified across investors. The social component matters only insofar as $r(\phi)$ is concentrated. If social attention is uniform, $R = 1/n$ and $H = 1/n$ for every ρ.

3 Network Illustrations

Figures 1–3 illustrate the objects in the model. Node size represents social centrality. The parameter ϕ governs how strongly differences in node size are translated into attention weights.

Figure 1: Decentralized ring network



Note: Centrality is nearly uniform, so centrality-based attention remains diversified and $R(\phi)$ stays close to its lower bound $1/n$.

In a decentralized ring, centralities are similar. Even when investors use a social feed, amplification cannot create a large concentration wedge because there is no dominant node. By contrast, in a hub-and-spoke network one node is substantially more central, so increasing ϕ shifts social attention toward the hub and increases $R(\phi)$.

Figure 3 anticipates the heterogeneous-precision extension. Concentration is not intrinsically inefficient. It is costly when the sources that receive high visibility are not correspondingly precise; it can be close to efficient when centrality and precision are aligned.

4 Centrality Crowding

4.1 Centrality amplification and price discovery

The central comparative static follows from the power transformation in Equation (3).

Proposition 2 (Centrality amplification crowds price weights)**.** *Suppose the centrality vector c is non-uniform and $\rho > 0$. Then $M(\rho, \phi)$ is strictly increasing in ϕ on $[0, \infty)$ in the sense that*

$$M(\rho, \phi_2) > M(\rho, \phi_1) \quad \textit{for every } \phi_2 > \phi_1 \geq 0.$$

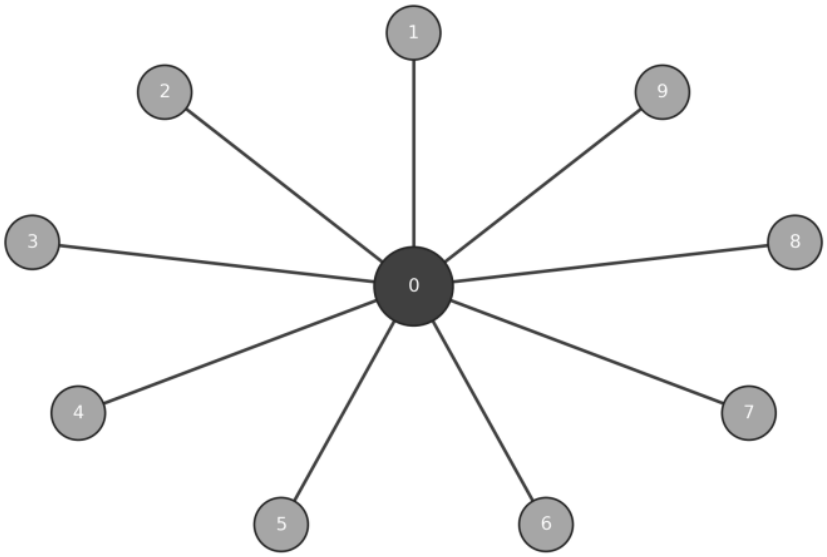
Moreover,

$$\frac{\partial M(\rho, \phi)}{\partial \phi} > 0 \quad \textit{for } \phi > 0, \qquad \left.\frac{\partial M(\rho, \phi)}{\partial \phi}\right|_{\phi=0} = 0. \tag{11}$$

Proof. See Appendix. □

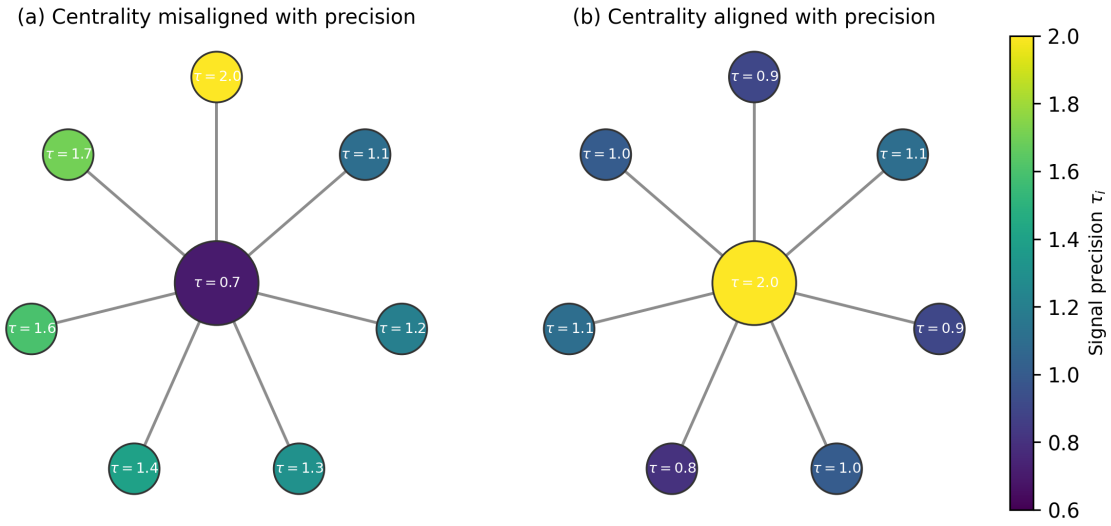
In this statement, ϕ_1 and ϕ_2 denote two alternative levels of the same centrality-amplification parameter, holding fixed the network c, social-reliance parameter ρ, and signal-noise variance σ^2. The boundary qualification at $\phi = 0$ is useful. Neutral attention is the minimum-concentration

Figure 2: Hub-and-spoke network



Note: Heterogeneous centrality allows the feed to concentrate attention on the hub as ϕ rises.

Figure 3: Centrality–precision alignment



Note: Node size represents social centrality and color represents signal precision. Panel (a) illustrates a large centrality wedge; panel (b) illustrates alignment between visibility and accuracy.

point, so an infinitesimal perturbation has zero first derivative there even though any positive amplification raises concentration relative to the neutral benchmark. Once $\phi > 0$, further amplification strictly increases the Herfindahl index of social attention and therefore M.

The economic mechanism is social redundancy. Because the signals are truthful, the loss of informativeness does not come from false information. Instead, market beliefs share the same centrality-weighted component. Raising ϕ makes that component more concentrated, so a smaller effective set of independent signal errors survives into the price.

4.2 One-period Gaussian correction

Because $\widetilde{p} = \theta + \eta$ with $\eta \sim N(0, M)$ independent of θ, where $\eta \equiv \sum_i a_i \varepsilon_i$ is aggregate pricing noise, standard Gaussian projection gives

$$\mathbb{E}[\theta \mid \widetilde{p}] = \frac{\sigma_\theta^2}{\sigma_\theta^2 + M} \widetilde{p}. \tag{12}$$

The coefficient $\sigma_\theta^2/(\sigma_\theta^2 + M) \in (0, 1)$ is the signal-extraction weight placed on the observed risk-adjusted price. It rises with fundamental variation σ_θ^2 and falls with pricing noise M: an observed price is more informative about θ when its variation is more heavily driven by fundamentals rather than aggregation error.

Hence the expected correction from the current risk-premium-adjusted price to the terminal payoff is

$$\mathbb{E}[\theta - \widetilde{p} \mid \widetilde{p}] = -\frac{M}{\sigma_\theta^2 + M} \widetilde{p}. \tag{13}$$

The coefficient $M/(\sigma_\theta^2 + M)$ is the noise share of the observed price variation. For a positive $\widetilde{p}$, the minus sign implies an expected downward correction; for a negative $\widetilde{p}$, it implies an expected upward correction. Its magnitude increases with M because a noisier price warrants a larger statistical correction toward the prior mean of the fundamental.

Corollary 1 (One-period projection correction)**.** *For a given positive realization $\widetilde{p} > 0$, the absolute expected correction in equation (13) is increasing in M. Under the equal-precision assumptions of Proposition 2, it is therefore strictly larger at higher ϕ whenever ϕ is increased in a non-uniform network with $\rho > 0$.*

Proof. See Appendix. □

Corollary 1 should not be read as a separate dynamic theory of return reversals. The model has one trading date and one terminal payoff date; equation (13) is a single signal-extraction projection. The statement that a noisier Gaussian price is expected to correct more toward the terminal fundamental is generic. The network-specific empirical restriction is that the noise term M is itself pinned down by observable attention concentration through $M = \sigma^2 H$, and that H moves systematically with centrality amplification ϕ.

4.3 Empirical interpretation

The sufficient-statistic formulation suggests a direct measurement strategy. Given an estimate of social attention shares r_i (from follower exposure, repost shares, feed impressions, or observed communication weights) and a measure of social reliance ρ, the model implies equilibrium price weights $a_i = (1 - \rho)/n + \rho r_i$ and concentration $H = \sum_i a_i^2$. In the equal-precision benchmark,

higher H predicts larger pricing-error variance and lower price informativeness. If a platform change or market event raises the amplification of central sources, the effect should be stronger where the underlying centrality vector is more unequal. The one-period correction coefficient $M/(\sigma_\theta^2 + M)$ should also be larger at higher H, but this implication is not unique to the network mechanism and therefore should be used jointly with network-based variation in H.

Section 6 adds the sharper cross-sectional prediction: the adverse relation between concentration and price discovery should weaken, disappear, or potentially reverse when highly central sources are also the most precise. Thus the empirical object is not centrality alone, but centrality concentration relative to source accuracy.

5 Social Attention and Network Diversification

The model separates the intensity of social attention, ρ, from the concentration of that attention, $R(\phi)$. Differentiating equation (9) gives

$$\frac{\partial H(\rho, \phi)}{\partial \rho} = 2\rho \left(R(\phi) - \frac{1}{n} \right). \tag{14}$$

The derivative has two economically distinct components. The factor ρ captures how much investors already rely on the social feed, while $R(\phi) - 1/n$ measures the feed's concentration above the equal-attention benchmark. Thus additional social reliance has no pricing-error effect when attention is diversified, but becomes increasingly costly as both reliance and concentration rise.

Proposition 3 (Social attention is harmful only when it is concentrated)**.** *If $r_i(\phi) = 1/n$ for all i, changing ρ has no effect on H or M. If $R(\phi) > 1/n$ and $\rho > 0$, then increasing ρ strictly increases H and M.*

Proof. See Appendix. □

This distinction prevents an overbroad interpretation of the model. Communication is not inefficient by assumption. A decentralized social statistic can preserve the same effective diversification as equal weighting. The loss arises when social reliance and attention concentration interact.

6 Precision, Optimal Attention, and the Centrality Wedge

6.1 Heterogeneous signal precision

The baseline assumes equal signal precision. Now let

$$x_i = \theta + \varepsilon_i, \qquad \mathrm{Var}(\varepsilon_i) = \tau_i^{-1}, \qquad \tau_i > 0, \tag{15}$$

where $\tau_i > 0$ is investor i's signal precision, defined as the inverse of her signal-error variance. Thus larger τ_i means a more accurate signal and smaller τ_i^{-1}. The equal-precision benchmark is nested as $\tau_i = 1/\sigma^2$ for every i. To isolate the allocation of attention across sources, consider the pure social-statistic case $\rho = 1$. For any feasible attention vector $r = (r_1, \ldots, r_n)$ satisfying

$r_i \geq 0$ and $\sum_i r_i = 1$, pricing-error variance is

$$M(r) = \sum_{i=1}^{n} \frac{r_i^2}{\tau_i}. \tag{16}$$

Each term r_i^2/τ_i is source i's contribution to pricing-error variance: a larger attention share r_i raises its contribution quadratically, while a larger precision τ_i reduces it. The notation $M(r)$ emphasizes that, in this extension, pricing error depends on the full allocation of attention rather than only on an unweighted Herfindahl index.

Proposition 4 (Precision alignment characterizes efficient attention)**.** *The unique attention vector minimizing $M(r)$ subject to $r_i \geq 0$ and $\sum_i r_i = 1$ is*

$$r_i^e = \frac{\tau_i}{\sum_{j=1}^{n} \tau_j}. \tag{17}$$

The superscript e denotes the efficient allocation. The denominator $\sum_j \tau_j$ is total signal precision in the market, so r_i^e assigns each investor an attention share equal to her share of total precision. Only relative precision determines these efficient weights; multiplying every τ_i by the same positive constant leaves r_i^e unchanged.

For any feasible r,

$$M(r) - M(r^e) = \sum_{i=1}^{n} \frac{(r_i - r_i^e)^2}{\tau_i} \geq 0. \tag{18}$$

The left-hand side is the excess pricing-error variance relative to the efficient benchmark. The right-hand side is a precision-weighted squared distance between actual and efficient attention: deviations from r_i^e are especially costly when they concern low-precision sources because $1/\tau_i$ is large. Consequently, centrality-based attention $r_i(\phi)$ is efficient if and only if

$$r_i(\phi) = r_i^e \quad \textit{for all } i, \tag{19}$$

which is equivalent to the precision-alignment condition

$$\tau_i = K c_i^{\phi} \quad \textit{for some common } K > 0. \tag{20}$$

Here K is a common positive scale factor. It has no effect on relative efficient attention; the condition requires only that cross-sectional differences in precision be proportional to the same centrality transformation c_i^{ϕ} that determines visibility. Thus efficiency requires a precise matching of relative *informational quality and* relative *social prominence.*

Proof. See Appendix. □

Equation (18) gives an exact measure of the centrality wedge. Define

$$\omega_i(\phi) = r_i(\phi) - r_i^e. \tag{21}$$

The variable $\omega_i(\phi)$ is investor i's attention misallocation relative to the precision-efficient benchmark. A positive ω_i means that investor i receives more social attention than her precision justifies; a negative value means that she is underweighted. The total efficiency loss is the precision-weighted squared distance $\sum_i \omega_i^2/\tau_i$.

Proposition 4 also qualifies Proposition 2. Under equal precision, any movement away from equal weights increases error, so centrality amplification is unambiguously harmful. With heterogeneous precision, however, concentration itself is not sufficient to sign the welfare effect. Amplification can improve price discovery if it moves attention toward the precision-efficient vector, and it worsens price discovery if it moves attention away from that vector. The empirically relevant condition is therefore alignment between social visibility and informational accuracy, not centrality concentration in isolation.

This distinction is especially useful for platform settings. A ranking system that amplifies central accounts need not be inefficient if centrality is a good proxy for forecast accuracy. If centrality instead reflects charisma, engagement, or legacy visibility, equation (18) predicts a measurable loss from misallocation of attention.

7 Conclusion

This paper presents a one-period limited-attention model in which network centrality shapes the weights that markets place on private signals. The analysis is intentionally behavioral rather than rational-expectations: investors do not learn from prices, residual perceived risk is exogenous, and net supply is fixed. These assumptions yield a transparent sufficient statistic. In the equal-precision benchmark, mean squared pricing error is $M = \sigma^2 H$, where H is the Herfindahl concentration of equilibrium signal weights. Centrality amplification increases H in non-uniform networks, providing a direct link from social visibility to weaker price discovery.

The Gaussian correction result is best viewed as an interpretation of this pricing noise, not as a separate intertemporal mechanism. The central empirical predictions are instead network-specific: attention concentration should forecast lower price informativeness when amplification increases the weight of already central nodes, and the effect should be attenuated when social centrality is aligned with signal precision. With heterogeneous precision, the exact efficiency condition is $r_i(\phi) \propto \tau_i$, or equivalently $\tau_i \propto c_i^{\phi}$. This alignment criterion provides a simple way to distinguish socially prominent information from informationally valuable information.

For empirical work, the model suggests combining network measures with observed exposure or engagement weights and ex post measures of source accuracy. For platform design, it suggests that preserving source diversity is valuable when visibility is weakly related to accuracy, whereas accuracy-aligned amplification can be benign. Extending the framework to endogenous price learning, stochastic supply, and genuinely dynamic trading would be natural directions for future research, but those mechanisms are intentionally outside the scope of the present one-period model.

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Appendix: Proofs

Proof of Proposition 1. Market clearing requires $\sum_{i=1}^{n} d_i = Q$. Substituting the individual demand equations yields:

$$\sum_{i=1}^{n} \frac{b_i - p}{\gamma \sigma_u^2} = Q$$
$$\sum_{i=1}^{n} (b_i - p) = \gamma \sigma_u^2 Q$$
$$np = \sum_{i=1}^{n} b_i - \gamma \sigma_u^2 Q$$
$$p = \frac{1}{n} \sum_{i=1}^{n} b_i - \frac{\gamma \sigma_u^2}{n} Q.$$

Substitute the belief formation rule $b_i = (1-\rho)x_i + \rho \sum_{j=1}^{n} r_j(\phi) x_j$ into the aggregate belief term:

$$\frac{1}{n} \sum_{i=1}^{n} b_i = \frac{1}{n} \sum_{i=1}^{n} \left[(1-\rho)x_i + \rho \sum_{j=1}^{n} r_j(\phi) x_j \right]$$
$$= \frac{1-\rho}{n} \sum_{i=1}^{n} x_i + \frac{\rho}{n} \sum_{i=1}^{n} \sum_{j=1}^{n} r_j(\phi) x_j.$$

Because the inner sum $\sum_{j=1}^{n} r_j(\phi) x_j$ is independent of i, summing it n times yields $n \sum_{j=1}^{n} r_j(\phi) x_j$. Thus:

$$\frac{1}{n} \sum_{i=1}^{n} b_i = \frac{1-\rho}{n} \sum_{i=1}^{n} x_i + \rho \sum_{j=1}^{n} r_j(\phi) x_j.$$

Relabeling the index j to i in the second term allows factoring out x_i:

$$\frac{1}{n} \sum_{i=1}^{n} b_i = \sum_{i=1}^{n} \left(\frac{1-\rho}{n} + \rho r_i(\phi) \right) x_i.$$

Therefore, the price equation is:

$$p = \sum_{i=1}^{n} a_i(\rho, \phi) x_i - \lambda Q,$$

where $a_i(\rho, \phi) = \frac{1-\rho}{n} + \rho r_i(\phi)$ and $\lambda = \frac{\gamma \sigma_u^2}{n}$.

To verify $\sum_{i=1}^{n} a_i = 1$:

$$\sum_{i=1}^{n} a_i = \sum_{i=1}^{n} \frac{1-\rho}{n} + \rho \sum_{i=1}^{n} r_i(\phi) = (1-\rho) + \rho(1) = 1.$$

The risk-adjusted price is $\widetilde{p} = p + \lambda Q = \sum_{i=1}^{n} a_i x_i$. Substituting $x_i = \theta + \varepsilon_i$:

$$\widetilde{p} = \sum_{i=1}^{n} a_i(\theta + \varepsilon_i) = \theta \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} a_i \varepsilon_i = \theta + \sum_{i=1}^{n} a_i \varepsilon_i.$$

The pricing error is $\widetilde{p} - \theta = \sum_{i=1}^{n} a_i \varepsilon_i$. Because $\varepsilon_i \overset{iid}{\sim} N(0, \sigma^2)$, the mean squared pricing error is the variance of this linear combination:

$$M(\rho, \phi) = \mathrm{Var}\left(\sum_{i=1}^{n} a_i \varepsilon_i\right) = \sum_{i=1}^{n} a_i^2 \mathrm{Var}(\varepsilon_i) = \sigma^2 \sum_{i=1}^{n} a_i^2 = \sigma^2 H(\rho, \phi).$$

Expanding the Herfindahl index of the price weights $H(\rho, \phi) = \sum_{i=1}^{n} a_i^2$:

$$\begin{aligned}
\sum_{i=1}^{n} \left(\frac{1-\rho}{n} + \rho r_i(\phi)\right)^2 &= \sum_{i=1}^{n} \left[\frac{(1-\rho)^2}{n^2} + \frac{2\rho(1-\rho)}{n} r_i(\phi) + \rho^2 r_i(\phi)^2\right] \\
&= \frac{(1-\rho)^2}{n} + \frac{2\rho(1-\rho)}{n} \sum_{i=1}^{n} r_i(\phi) + \rho^2 \sum_{i=1}^{n} r_i(\phi)^2 \\
&= \frac{1 - 2\rho + \rho^2 + 2\rho - 2\rho^2}{n} + \rho^2 R(\phi) \\
&= \frac{1-\rho^2}{n} + \rho^2 R(\phi) \\
&= \frac{1}{n} + \rho^2 \left(R(\phi) - \frac{1}{n}\right).
\end{aligned}$$

□

Proof of Proposition 2. Let $\ell_i = \log c_i$. Define the log-partition function $A(t) = \log\left(\sum_{i=1}^{n} e^{t\ell_i}\right)$. The social weight can be written as:

$$r_i(\phi) = \frac{e^{\phi \ell_i}}{\sum_{j=1}^{n} e^{\phi \ell_j}} = e^{\phi \ell_i - A(\phi)}.$$

The Herfindahl index of the social attention weights is:

$$R(\phi) = \sum_{i=1}^{n} r_i(\phi)^2 = \sum_{i=1}^{n} e^{2\phi \ell_i - 2A(\phi)} = e^{A(2\phi) - 2A(\phi)}.$$

Taking the natural logarithm and differentiating with respect to ϕ:

$$\begin{aligned}
\log R(\phi) &= A(2\phi) - 2A(\phi) \\
\frac{d}{d\phi} \log R(\phi) &= 2A'(2\phi) - 2A'(\phi).
\end{aligned}$$

The first derivative of the log-partition function $A(t)$ is the expectation of ℓ_i under the

exponentially tilted measure $P_t(\ell_i) \propto e^{t\ell_i}$:

$$A'(t) = \frac{\sum_{i=1}^n \ell_i e^{t\ell_i}}{\sum_{i=1}^n e^{t\ell_i}} = \mathbb{E}_t[\ell_i].$$

The second derivative $A''(t)$ is the variance of ℓ_i under this same measure:

$$\begin{aligned} A''(t) &= \frac{\left(\sum_{i=1}^n \ell_i^2 e^{t\ell_i}\right)\left(\sum_{i=1}^n e^{t\ell_i}\right) - \left(\sum_{i=1}^n \ell_i e^{t\ell_i}\right)^2}{\left(\sum_{i=1}^n e^{t\ell_i}\right)^2} \\ &= \mathbb{E}_t[\ell_i^2] - (\mathbb{E}_t[\ell_i])^2 = \mathrm{Var}_t(\ell_i). \end{aligned}$$

Because the centrality vector c is non-uniform, the elements ℓ_i are not all identical. Therefore, the variance under any strictly positive probability measure is strictly positive, meaning $A''(t) > 0$ for all $t \in \mathbb{R}$. This implies $A'(t)$ is a strictly monotonically increasing function.

For any $\phi > 0$, we have $2\phi > \phi$, which implies $A'(2\phi) > A'(\phi)$. Thus:

$$\frac{d}{d\phi} \log R(\phi) = 2[A'(2\phi) - A'(\phi)] > 0.$$

Since $R(\phi) > 0$, this implies $R'(\phi) > 0$ for $\phi > 0$. At $\phi = 0$, $2\phi = \phi = 0$, so $A'(0) - A'(0) = 0$, establishing $R'(0) = 0$.

Applying the chain rule to the expression for mean squared pricing error $M(\rho, \phi) = \sigma^2 \left[\frac{1}{n} + \rho^2 \left(R(\phi) - \frac{1}{n}\right)\right]$ gives:

$$\frac{\partial M}{\partial \phi} = \sigma^2 \rho^2 R'(\phi).$$

For $\rho > 0$, $\frac{\partial M}{\partial \phi} > 0$ when $\phi > 0$, and $\frac{\partial M}{\partial \phi} = 0$ when $\phi = 0$. Because the derivative is strictly positive for all $\phi > 0$, $M(\rho, \phi)$ is strictly increasing in ϕ on $[0, \infty)$. □

Proof of Corollary 1. From Proposition 1, $\widetilde{p} = \theta + \eta$, where $\eta = \sum_{i=1}^n a_i \varepsilon_i \sim N(0, M)$. Since $\theta \sim N(0, \sigma_\theta^2)$ and is independent of each ε_i, θ and η are independent.

$$\begin{aligned} \mathrm{Var}(\widetilde{p}) &= \mathrm{Var}(\theta + \eta) = \sigma_\theta^2 + M \\ \mathrm{Cov}(\theta, \widetilde{p}) &= \mathrm{Cov}(\theta, \theta + \eta) = \mathrm{Var}(\theta) = \sigma_\theta^2. \end{aligned}$$

By standard properties of joint Gaussian distributions, the conditional expectation is:

$$\mathbb{E}[\theta \mid \widetilde{p}] = \frac{\mathrm{Cov}(\theta, \widetilde{p})}{\mathrm{Var}(\widetilde{p})} \widetilde{p} = \frac{\sigma_\theta^2}{\sigma_\theta^2 + M} \widetilde{p}.$$

The expected correction from the current price toward the terminal payoff is:

$$\mathbb{E}[\theta - \widetilde{p} \mid \widetilde{p}] = \mathbb{E}[\theta \mid \widetilde{p}] - \widetilde{p} = \left(\frac{\sigma_\theta^2}{\sigma_\theta^2 + M} - 1\right) \widetilde{p} = -\frac{M}{\sigma_\theta^2 + M} \widetilde{p}.$$

For a fixed positive realization $\widetilde{p} > 0$, the absolute expected correction is $C(M) = \frac{M}{\sigma_\theta^2 + M}\widetilde{p}$. Differentiating with respect to M:

$$\frac{\partial C}{\partial M} = \widetilde{p}\left[\frac{1(\sigma_\theta^2 + M) - M(1)}{(\sigma_\theta^2 + M)^2}\right] = \widetilde{p}\frac{\sigma_\theta^2}{(\sigma_\theta^2 + M)^2} > 0.$$

Since $C(M)$ is strictly increasing in M, and Proposition 2 establishes that M is strictly increasing in ϕ (for non-uniform networks and $\rho > 0$), the absolute expected correction strictly increases at higher values of ϕ. □

Proof of Proposition 3. By the Cauchy-Schwarz inequality applied to the vectors $r = (r_1, \ldots, r_n)$ and $\mathbf{1} = (1, \ldots, 1)$:

$$\left(\sum_{i=1}^{n} r_i \cdot 1\right)^2 \leq \left(\sum_{i=1}^{n} r_i^2\right)\left(\sum_{i=1}^{n} 1^2\right).$$

Since $\sum_{i=1}^{n} r_i = 1$, this simplifies to:

$$1^2 \leq R(\phi) \cdot n \implies R(\phi) \geq \frac{1}{n}.$$

Equality holds if and only if r_i is proportional to 1, meaning $r_i = 1/n$ for all i.

If $r_i(\phi) = 1/n$ for all i, then $R(\phi) = 1/n$. The Herfindahl index of the price weights becomes $H(\rho, \phi) = \frac{1}{n} + \rho^2\left(\frac{1}{n} - \frac{1}{n}\right) = \frac{1}{n}$, and $M(\rho, \phi) = \frac{\sigma^2}{n}$. Both are constant with respect to ρ.

Differentiating $H(\rho, \phi)$ with respect to ρ:

$$\frac{\partial H}{\partial \rho} = 2\rho\left(R(\phi) - \frac{1}{n}\right).$$

If $R(\phi) > 1/n$ and $\rho > 0$, both terms in the product are strictly positive, making $\frac{\partial H}{\partial \rho} > 0$. Because $M = \sigma^2 H$, $\frac{\partial M}{\partial \rho}$ is identically strictly positive under these conditions. □

Proof of Proposition 4. The objective is to minimize $M(r) = \sum_{i=1}^{n} \frac{r_i^2}{\tau_i}$ subject to $\sum_{i=1}^{n} r_i = 1$. Form the Lagrangian:

$$\mathcal{L}(r, \mu) = \sum_{i=1}^{n} \frac{r_i^2}{\tau_i} - \mu\left(\sum_{i=1}^{n} r_i - 1\right).$$

Taking the partial derivative with respect to r_i and setting it to zero:

$$\frac{\partial \mathcal{L}}{\partial r_i} = \frac{2r_i}{\tau_i} - \mu = 0 \implies r_i = \frac{\mu}{2}\tau_i.$$

To find the multiplier μ, enforce the summation constraint:

$$\sum_{i=1}^{n} r_i = \sum_{i=1}^{n} \frac{\mu}{2}\tau_i = 1 \implies \frac{\mu}{2}\sum_{i=1}^{n}\tau_i = 1 \implies \frac{\mu}{2} = \frac{1}{\sum_{j=1}^{n}\tau_j}.$$

Substituting this back gives the efficient attention weights:

$$r_i^e = \frac{\tau_i}{\sum_{j=1}^{n}\tau_j}.$$

Since $M(r)$ is a strictly convex function (a sum of quadratic terms with strictly positive coefficients $1/\tau_i$), this critical point is the unique global minimum.

To evaluate $M(r) - M(r^e)$, expand the precision-weighted squared distance between an arbitrary feasible vector r and the efficient vector r^e:

$$\sum_{i=1}^{n}\frac{(r_i - r_i^e)^2}{\tau_i} = \sum_{i=1}^{n}\frac{r_i^2}{\tau_i} - 2\sum_{i=1}^{n}\frac{r_i r_i^e}{\tau_i} + \sum_{i=1}^{n}\frac{(r_i^e)^2}{\tau_i}.$$

Notice that $\frac{r_i^e}{\tau_i} = \frac{1}{\sum_{j=1}^{n}\tau_j}$ is a constant for all i. Let $T = \sum_{j=1}^{n}\tau_j$. The middle sum simplifies to:

$$-2\sum_{i=1}^{n} r_i\left(\frac{1}{T}\right) = -\frac{2}{T}\sum_{i=1}^{n} r_i = -\frac{2}{T}.$$

The final sum evaluates to the efficient pricing error $M(r^e)$:

$$M(r^e) = \sum_{i=1}^{n}\frac{(r_i^e)^2}{\tau_i} = \sum_{i=1}^{n} r_i^e\left(\frac{1}{T}\right) = \frac{1}{T}\sum_{i=1}^{n} r_i^e = \frac{1}{T}.$$

Therefore:

$$\sum_{i=1}^{n}\frac{(r_i - r_i^e)^2}{\tau_i} = M(r) - \frac{2}{T} + \frac{1}{T} = M(r) - M(r^e).$$

Because the sum of squares is non-negative, $M(r) - M(r^e) \geq 0$.

Finally, centrality-based attention is efficient if and only if $r_i(\phi) = r_i^e$ for all i:

$$\frac{c_i^\phi}{\sum_{j=1}^{n} c_j^\phi} = \frac{\tau_i}{\sum_{j=1}^{n}\tau_j}.$$

Solving for τ_i yields $\tau_i = c_i^\phi \frac{\sum_{j=1}^{n}\tau_j}{\sum_{j=1}^{n} c_j^\phi}$. Setting $K = \frac{\sum_{j=1}^{n}\tau_j}{\sum_{j=1}^{n} c_j^\phi} > 0$ confirms the exact precision-alignment condition $\tau_i = K c_i^\phi$. □